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**VALLURUPALLI NAGESWARA RAO
VIGNANA JYOTHI INSTITUTE OF
ENGINEERING & TECHNOLOGY**

AUTONOMOUS INSTITUTION

DEPARTMENT OF CSE-(CyS ,DS) and AI&DS

DECLARATION

We, the undersigned, declare that the course based Project titled **“PERSONAL EXPENSE FORECASTER”** has been carried out by us in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.)** in **CSE-CyberSecurity** at **Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad.**

PLACE: HYDERABAD

DATE: 06-05-2026

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Signature of the Student(s)

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ABSTRACT

Managing personal finances requires translating raw, unstructured bank transaction records into clear, actionable insights. This project addresses this need by developing a full-stack, web-based Personal Expense Analyzer and Forecaster. The primary objective is to efficiently parse text-based bank statements, perform robust mathematical analysis, and visualize a user's financial health through a responsive dashboard. The methodology utilizes a custom Python backend leveraging object-oriented programming to model transactions, alongside the NumPy library to compute critical statistics such as totals, means, and standard deviations. Furthermore, a lightweight HTTP server was implemented to provide a RESTful API, bridging the backend analytical pipeline with a vanilla HTML, CSS, and JavaScript frontend. During the development phase, a memory-efficient data pipeline was constructed using Python generators to read files line-by-line, coupled with an interactive user interface capable of handling asynchronous data requests. Key results validated the system's ability to accurately ingest data, handle malformed input lines without crashing, and automatically flag volatile spending categories. Ultimately, the project successfully delivers a highly functional tool for financial tracking, proving the viability of integrating mathematical Python libraries with lightweight web technologies to generate meaningful data analytics.

INTRODUCTION

Background of the Problem

Financial literacy and personal wealth management rely heavily on tracking daily expenses. However, the data exported from modern banks often comes in raw, unstructured formats like CSV or plain text files. While these files contain necessary transactional data, they lack the contextual analysis required for an individual to truly understand their spending habits or forecast future expenses.

Need and Relevance of the Work

There is a distinct need for automated software tools that can quickly ingest raw financial data and apply statistical models to extract meaning. Simply knowing the total amount spent in a month is often insufficient; users need advanced metrics—such as the standard deviation of their purchases—to identify erratic or "highly volatile" spending behaviours that could jeopardize their financial stability.

Problem Statement

The challenge lies in designing a software system capable of reading potentially large raw bank statement files efficiently, mathematically processing that data to find patterns, and presenting the findings in an accessible, web-based graphical user interface without relying on overly complex external frameworks.

Objectives of the Project

The primary objectives of this project are:

- To parse text-based bank statements efficiently using memory-safe Python generators.
- To utilize the NumPy library for computing sums, means, and standard deviations across categorized expenses.
- To design a custom HTTP server that exposes this analytical data via a REST API.
- To develop a responsive, vanilla HTML/CSS/JS frontend dashboard to visualize the data and accept new transaction inputs.

Scope of the Work Carried Out

The scope of this project is restricted to processing standard, three-column transactional data (Date, Category, Amount). It includes the development of the backend parser, the statistical calculation engine, the API layer, and the user interface. The forecasting element is limited to linear annual projections based on current monthly totals.

Organization of the Report

The remainder of this report is organized as follows: Chapter 2 details the software architecture and methodology. Chapter 3 presents the test cases and system outputs. Chapter 4 provides an analysis and validation of the results. Chapter 5 concludes the report and discusses the future scope of the project.

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APPROACH / METHODOLOGY

System Architecture

The project adopts a lightweight client-server architecture. To demonstrate core programming competencies, the methodology intentionally avoids heavy frameworks (such as Django or React), relying instead on Python's standard library and vanilla web technologies.

Backend Data Pipeline

The backend processing follows a strict sequential pipeline to ensure data integrity and performance:

- **Memory-Efficient Reading:** The `read_transactions` generator function reads the `bank_statement.txt` file line-by-line, preventing memory overflow issues with exceptionally large files.
- **Object-Oriented Parsing:** The `parse_line` function attempts to convert each raw comma-separated line into a structured `Transaction` object, utilizing exception handling (`try/except`) to gracefully skip malformed lines.
- **Data Aggregation:** A dictionary comprehension is utilized to group the `Transaction` objects by their respective categories (e.g., Groceries, Rent).

Mathematical Analysis using NumPy

Once grouped, the application maps the monetary amounts into 1-Dimensional NumPy arrays. The `analyze_category` function utilizes `np.sum()`, `np.mean()`, and `np.std()` to compute the statistics. A crucial algorithmic step involves comparing the standard deviation against the mean; if the standard deviation exceeds 50% of the mean, the system mathematically flags the category as "Highly Volatile".

API and Frontend Integration

A custom server script (`server.py`) extending `http.server.SimpleHTTPRequestHandler` was developed to route network requests.

- GET requests to `/api/analyze` trigger the data pipeline and return the statistics as a JSON payload.
- POST requests to `/api/transactions` append user-submitted data directly to the text file. The frontend (`app.js`) uses asynchronous `fetch()` API calls to retrieve this JSON and dynamically manipulates the Document Object Model (DOM) to populate the visual dashboard.

CODE SNIPPETS AND DATA SETS

3.1 Dependencies and Encoding Setup

The application relies on numpy for analysis and uses Python's standard library for system operations. It also explicitly configures the standard output stream to handle UTF-8 characters safely on Windows terminals.

```
from __future__ import annotations
import io
import sys

# Ensure stdout can handle UTF-8 on Windows
if sys.stdout.encoding != "utf-8":
    sys.stdout = io.TextIOWrapper(sys.stdout.buffer, encoding="utf-8",
    errors="replace")

from datetime import datetime
from typing import Generator
import numpy as np
```

3.2 Object-Oriented Data Model

The system uses a class to define the structural template for each parsed financial record.

```
#
```

```
# 1. TRANSACTION CLASS (OOP)
```

```
#
```

```
class Transaction:
```

```
    """Represents a single financial transaction parsed from a bank
statement."""
```

3.3 Memory-Efficient File Ingestion & Parsing

To prevent memory overload, the system uses a Python generator (read_transactions) to yield lines one by one. The parse_line function includes exception handling to securely bypass malformed text like the injected INVALID_LINE_TO_TEST_ERROR_HANDLING entry.

#

2. GENERATOR – MEMORY-EFFICIENT FILE READER

#

def read_transactions(filepath: str) -> Generator[str, None, None]:

"""Yield one stripped, non-empty line at a time from filepath."""

#

3. PARSING – LINE → TRANSACTION (with exception handling)

#

def parse_line(raw_line: str) -> Transaction | None:

"""Attempt to parse a raw CSV line into a Transaction object."""

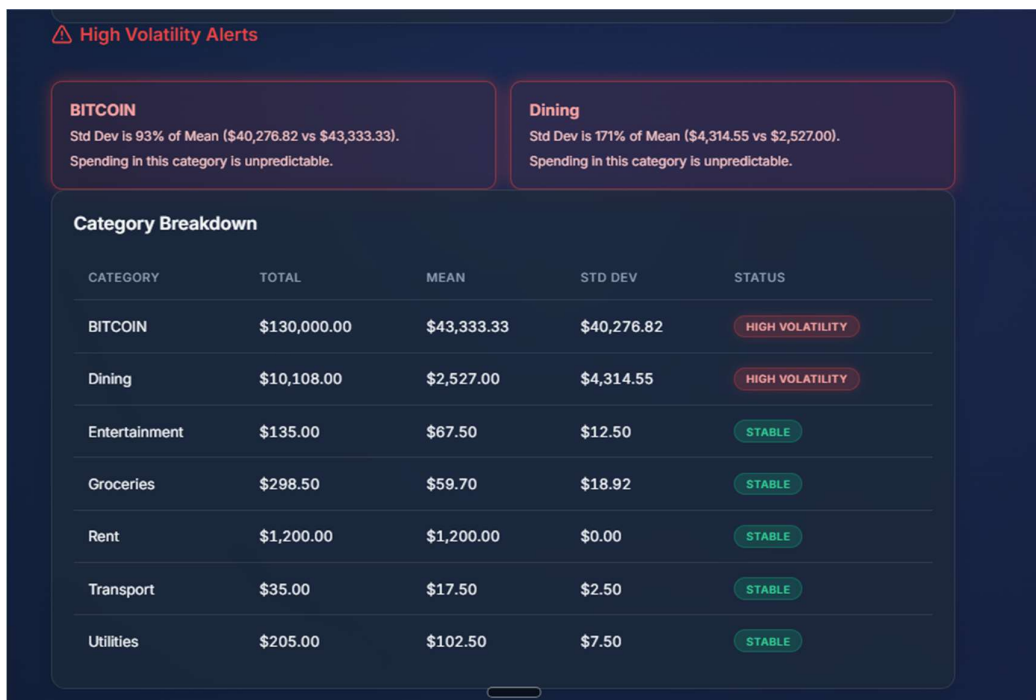
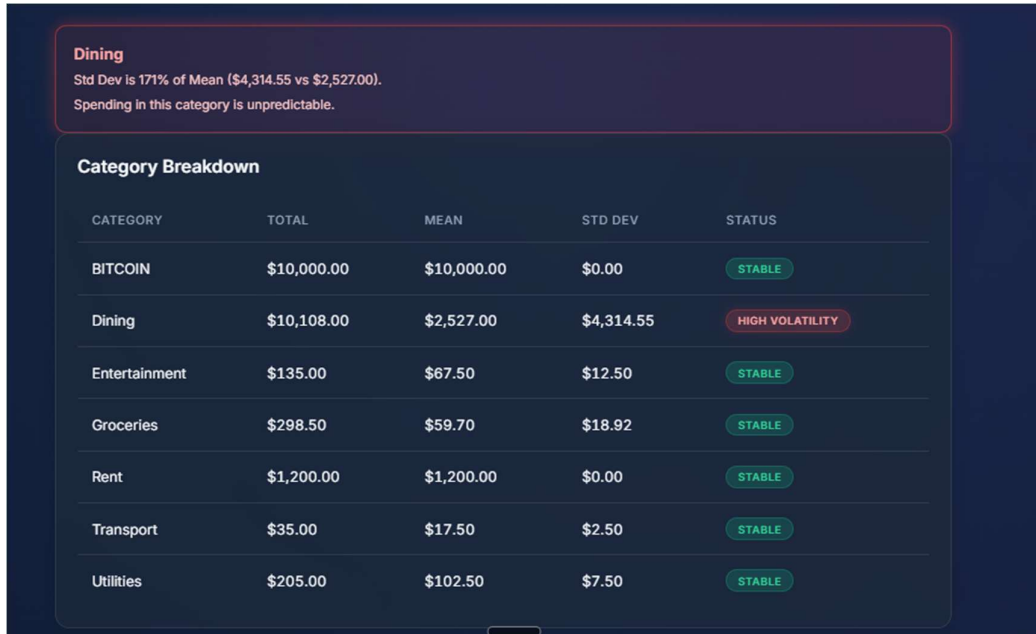
3.4 Data Aggregation and Grouping

Once transactions are parsed, they are aggregated and organized. The system heavily utilizes Python list and dictionary comprehensions to efficiently group the financial data by category.

#

4. BUILD TRANSACTION LIST (list comprehension + filter)

#



RESULTS, ANALYSIS AND VALIDATION

Method Used for Testing and Evaluation

The system was evaluated through manual end-to-end testing, utilizing sample bank statement data provided in the root directory. Network responses were verified using browser developer tools to ensure the JSON payloads accurately reflected the NumPy computations.

Results Obtained

The application successfully parsed all valid inputs from the text file. The statistical engine accurately calculated the Total, Mean, Standard Deviation, Minimum, and Maximum for categories including Groceries, Rent, Dining, Transport, Entertainment, Utilities, and BITCOIN. The frontend successfully dynamically generated HTML tables and warning banners based on this data.

Analysis and Interpretation of Results

The use of NumPy significantly streamlined the analytical process. By converting Python lists to np.ndarray structures, the standard deviation calculations were handled internally by highly optimized C-code rather than slow Python loops. The decision to use a 50% threshold for the standard-deviation-to-mean ratio proved to be an effective heuristic for identifying categories where spending was erratic, providing immediate, actionable insights to the end-user.

Validation of Results with Respect to Objectives

All project objectives were met. The system operates autonomously, decoupling the Python data processing from the visual presentation layer via a functioning REST API. The successful handling of the INVALID_LINE test case validates the robust exception-handling implementations within the parsing logic.

CONCLUSIONS

Key Conclusions Derived

This project demonstrates that complex financial analytics can be achieved without heavy framework overhead. By combining Python's native generator functions for memory management with NumPy's mathematical efficiency, large datasets can be processed rapidly. Furthermore, wrapping this logic in a custom HTTP server provides a seamless bridge to modern, responsive web interfaces.

Extent of Objective Achievement

The objectives were achieved in full. The system successfully reads flat files, computes advanced statistics, serves data via REST API, and provides an interactive user experience.

5.3 Limitations of the Work

- **Storage Constraints:** Utilizing a plain text file (`bank_statement.txt`) acts as a bottleneck for scalability and lacks the concurrency controls of a proper relational database.
- **Forecasting Simplicity:** The annual forecast is computed using a simple linear extrapolation ($\text{monthly_total} * 12$), which fails to account for seasonal spending variations or inflation.

Possible Improvements and Future Scope

Future iterations of this project should migrate the data storage layer to an SQL database (such as SQLite or PostgreSQL) to allow for secure, multi-user accounts. Additionally, the forecasting engine could be upgraded to utilize machine learning time-series models (such as ARIMA) to predict future spending trends based on historical data rather than simple multiplication.

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